Three data-generating architectures for biomarker-treatment interactions in aggregated N-of-1 trials, and why the choice changes power under carryover: a tutorial

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1 Abstract

Background. Simulation is the standard tool for sizing aggregated N-of-1 trials – designs in which each participant is repeatedly crossed on and off treatment and many single-patient series are pooled – intended to validate predictive biomarkers. Any such simulation must encode how the biomarker moderates the treatment effect in its data-generating process (DGP), and the encoding is a substantive choice. This tutorial, aimed at applied statisticians who may be new to the N-of-1 setting, formalizes *two* architectures and shows that the choice between them, usually treated as an implementation detail, materially changes the estimated power loss under carryover.

Methods. We define two DGP architectures. *The mean-moderation architecture* (direct mean moderation): the biomarker scales the treatment effect in the conditional mean. *The covariance-moderation architecture* (differential correlation, the multivariate normal (MVN) approach): the interaction is encoded as treatment-state-dependent correlation in the joint distribution. Holding the analysis model fixed (a linear mixed model with continuous-time AR(1) residual correlation), we quantify by Monte Carlo simulation the power to detect the interaction as carryover increases, across three within-subject designs, calibrated to a prazosin/PTSD reference application.

Results. Under the mean-moderation architecture power is nearly preserved because the biomarker-effect proportionality is maintained at every occasion: at the reference cell it falls only from 0.751 to 0.730 (2.8% relative) as the carryover half-life grows to one week. Under the covariance-moderation architecture the same carryover erodes the differential correlation that carries the signal, and power falls from 0.777 to 0.539 (30.6% relative), roughly eleven times the mean-moderation architecture loss and far beyond Monte Carlo error ($z = 7.64$, $p < 10^{-13}$). The discrepancy is strongly design-dependent: large in designs with a discontinuation phase, negligible in a classical crossover whose within-subject AR(1) structure already dominates the correlation channel.

Conclusions. The DGP architecture is a substantive scientific assumption, not a parameterization convenience: it determines how severely carryover appears to reduce power. We recommend that biomarker-validation simulation studies state which of the two architectures they adopt and, when the mechanism is uncertain, report power under both.

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2 1. Introduction

[1]

Power for these designs is estimated by simulation, and the simulation must encode the biomarker interaction in its DGP – in more than one possible way.

- The target is the biomarker-treatment interaction.
- Closed-form power exists only in simplified cases [@pmsimstats-paper08]; for the designs and model here it is obtained by Monte Carlo simulation.
- How the interaction is built into the DGP is a modeling choice, the subject of this paper.

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Precision-medicine trials increasingly ask whether a baseline **biomarker** modifies the treatment effect – a biomarker-treatment interaction. For simplified special cases, such as a balanced strict repeated-measures analysis of variance (RM-ANOVA) under sphericity, the power of the interaction test does have a closed form, and a companion paper[1] derives it together with its equivalence to a two-sample *t*-test. For the continuous-time mixed model and the N-of-1 designs considered here, however, no such closed form is available, so power is estimated by Monte Carlo simulation: many trials are generated, analyzed, and the rejection rate recorded. The point that motivates this paper is that the simulation must itself encode the interaction in its data-generating process, and there is more than one way to do so.

[2]

There are two architectures: mean moderation and differential correlation.

- The mean-moderation architecture: an interaction term scales the treatment effect in the mean; standard across trial-simulation frameworks.
- The covariance-moderation architecture: the biomarker changes the response-outcome correlation, encoding the interaction in the joint distribution (Hendrickson's N-of-1 approach).
- A dual-channel architecture activating both at once is developed in a companion paper (Section 2.2).

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We distinguish two architectures. The common one, **direct mean moderation** (the mean-moderation architecture), adds an interaction term to the outcome equation so the biomarker scales the size of the treatment effect; essentially every standard trial-simulation framework uses it[2,3]. The second, **differential correlation** (the covariance-moderation architecture), used by Hendrickson et al.[4] for N-of-1 trials, instead makes the biomarker change the correlation between treatment response and outcome, so the interaction is encoded in the joint distribution rather than in the mean; on this reading it is the reduced-form second-moment encoding of a latent responder-class mechanism (Section 4;[5]). We treat the two as co-equal data-generating architectures, defined precisely in Section 2. A third, **combined** architecture, which activates both channels simultaneously through two independent parameters and recovers these two as its single-channel limits, is developed in a companion paper[6].

In plain language, without the latent-variable vocabulary: suppose patients come in two kinds, responders and non-responders, but we cannot see which kind a given patient is. The biomarker is only an imperfect clue to that hidden status, so two patients with the same biomarker value can still respond differently. The mean-moderation architecture cannot represent this, because it assumes the biomarker fixes each patient's drug effect exactly; the covariance-moderation architecture represents it by letting biomarker and response track each other only while the drug acts, as a tendency rather than a fixed rule. That tendency fades once the drug washes out, which is why carryover costs more power under the covariance-moderation architecture than under the mean-moderation architecture (Section 4 gives clinical examples).

[3]

The two encodings, found in two implementations of the same software, gave different carryover stories; this paper explains why.

- The original code used the covariance-moderation architecture; a re-implementation used the mean-moderation architecture.
- The two disagreed about how much carryover hurts power.
- We formalize both and give guidance on which to use.

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We ran into both while building the pmsimstats software: the original code used the covariance-moderation architecture, while a re-implementation had quietly switched to the mean-moderation architecture, and the two gave different stories about carryover. This paper pins down why. We define the two architectures precisely, show by simulation that they react very differently to carryover, and give simple guidance on which to use.

[4]

Primer: aggregated N-of-1 trials pool repeated within-patient on/off contrasts, and carryover is their distinguishing nuisance.

- A single N-of-1 trial repeatedly crosses one patient on/off; pooling many gives population inference at small N.
- Carryover is residual drug effect decaying over days to weeks, contaminating off-drug occasions.
- Three within-subject designs (crossover (CO), open-label with blinded discontinuation (OL+BDC), Hybrid) differ in how on/off occasions are arranged.

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The N-of-1 setting, for readers new to it. In a single N-of-1 trial one participant is repeatedly crossed between active treatment and a comparator over a sequence of measurement occasions, so the within-participant on-versus-off contrast estimates that participant's individual treatment effect; an *aggregated* N-of-1 trial pools many such series in a mixed model to obtain population-level inference at far smaller sample sizes than a parallel-group design. The feature that distinguishes these designs analytically is *carryover*: residual drug effect that decays over days to weeks after discontinuation, contaminating off-drug occasions. We study three within-subject designs that differ in how on- and off-drug occasions are arranged – a traditional crossover (CO), an open-label-plus-blinded- discontinuation design (OL+BDC), and a Hybrid N-of-1 design – defined in Section 3.

3 2. Two DGP architectures

[5]

Notation: B_i biomarker, D_{it} on-drug indicator, BR_{it} biological response, with `Dbc` the analysis-layer exposure.

- B_i, D_{it}, BR_{it} describe the data-generating layer.
- The analysis layer uses a continuous exposure `Dbc` (Section 3.1).
- Typewriter identifiers denote code/formula variables; math symbols are used otherwise.

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Notation. Throughout this paper, B_i denotes the biomarker value of participant i, D_{it} the binary on-drug indicator at timepoint t (1 on drug, 0 off drug), and BR_{it} the biological response component of the outcome. These three symbols describe the data-generating-process layer. The analysis-model layer uses a related but distinct continuous drug exposure variable, `Dbc`, defined in Section 3.1. Code-style identifiers in typewriter font (`bm`, `Dbc`, `BR`) refer to specific variables in the pmsimstats codebase or to R formula syntax; mathematical symbols are used otherwise.

3.1 2.1 The Mean-Moderation Architecture: Direct Mean Moderation

In this approach the biomarker enters the mean structure of the outcome directly. The treatment effect for participant i at timepoint t is:

$$Y_{it} = \mu_0 + \beta_1 \cdot D_{it} \cdot (1 + \beta_{bm} \cdot B_i) + \epsilon_{it}$$

[6]

Under the mean-moderation architecture the interaction is explicit in the mean: a higher biomarker gives a proportionally larger treatment effect.

- β_{bm} is the biomarker moderation parameter on the centered biomarker.
- The effect scales with B_i deterministically in the conditional mean.
- The signal lives in the first moment of the outcome.

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where D_{it} is the binary on-drug indicator (or a continuous measure of drug exposure), B_i is the participant's biomarker value (typically centered), and β_{bm} is the biomarker moderation parameter. The interaction is explicit in the mean: participants with higher biomarker values have proportionally larger treatment effects.

Key properties:

- The interaction is **deterministic** given the biomarker value and treatment status.
- The biomarker moderation β_{bm} is a **population-level parameter** that applies uniformly to all participants.

- The signal for the interaction is in the **first moment** (conditional mean) of the outcome distribution.
- Adding noise, random effects, or carryover to D_{it} scales the entire treatment effect (including its biomarker-dependent component) proportionally. The **ratio** of drug effect between high-biomarker and low-biomarker participants is preserved.

  [7]

  The multiplicative form maps onto the standard additive effect-modeling regression, so both are instances of the mean-moderation architecture.

  - The additive interaction β_3 equals $\beta_1 \cdot \beta_{bm}$ in the multiplicative form.
  - The additive form adds a prognostic main effect β_2 the multiplicative form sets to zero.
  - For interaction power the two are interchangeable up to that prognostic distinction.

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  We note that the multiplicative form above maps directly onto the standard additive regression form for predictive biomarker effects used in the trial methodology literature, specifically the effect modeling equation of Kent et al.[7]:

  $$Y_{it} = \beta_0 + \beta_1 D_{it} + \beta_2 B_i + \beta_3 D_{it} B_i + \epsilon_{it}$$

  The interaction coefficient β_3 in the additive form corresponds to the product $\beta_1 \cdot \beta_{bm}$ in the multiplicative form. The additive form additionally accommodates a prognostic main effect β_2 for the biomarker, which the multiplicative form sets to zero by construction. For the purpose of power calculations on the interaction coefficient, the two forms are interchangeable up to this prognostic-effect distinction, and both fall within the mean-moderation architecture.

This architecture is used in the pedagogical simulation (`implementations/simple/simulation.R`) and in the exploratory carryover analyses (`simulation_carryover_spectrum.R`) of the pmsimstats project.

3.2 2.2 The Covariance-Moderation Architecture: Differential Correlation (MVN)

In this approach the biomarker and the biological response (BR) component are jointly drawn from a multivariate normal distribution with treatment-state-dependent correlation:

$$\begin{pmatrix} B_i \\ BR_{it} \end{pmatrix} \sim \text{MVN}\left(\begin{pmatrix} \mu_B \\ \mu_{BR}(t, D_{it}) \end{pmatrix}, \Sigma(D_{it}) \right)$$

where the covariance matrix Σ has the following treatment-state-dependent BM-BR correlation:

$$\text{Cor}(B_i, BR_{it}) = \begin{cases} c_{bm} & \text{if } D_{it} = 1 \text{ (on drug)} \\ c_{bm} \cdot e^{-\lambda \cdot t_{sd}} & \text{if } D_{it} = 0 \text{ (off drug)} \end{cases}$$

where t_{sd} is time since drug discontinuation. The exponential-decay form is the default; the limiting cases $\lambda = 0$ (no decay; correlation persists indefinitely during the off-drug phase) and $\lambda \to \infty$

(immediate decay; correlation drops to zero the instant the drug is discontinued) recover the two no-carryover model variants.

The interaction is not in the mean structure: the population mean of BR_{it} is the same for all participants with the same treatment history. Instead it emerges from the **conditional distribution**: given a participant's biomarker value $B_i = b$, the conditional expectation of their biological response is:

$$E[BR_{it}|B_i = b, D_{it} = 1] = \mu_{BR}(t) + c_{bm} \cdot \frac{\sigma_{BR}}{\sigma_B} \cdot (b - \mu_B)$$

[8]

Under the covariance-moderation architecture the interaction is in the second moment: the biomarker modulates BR through a correlation that decays during the off-drug period.

- The conditional mean of BR rises with b via the correlation, not a mean parameter.
- Effective correlation is c_{bm} on-drug, $c_{bm}e^{-\lambda t_{sd}}$ off-drug.
- Moderation persists with attenuated strength off-drug and vanishes only at complete decay.

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This conditional expectation has an interaction-like structure: the biomarker value b modulates the expected BR. The effective correlation is c_{bm} on drug and $c_{bm} \cdot e^{-\lambda t_{sd}}$ during the off-drug period; the moderation therefore persists with attenuated strength during off-drug periods and vanishes only in the limit of complete decay.

Key properties:

- The interaction is **probabilistic**: it manifests as a tendency, not a deterministic scaling.
- The biomarker moderation c_{bm} is a **correlation parameter** that governs the strength of the association.
- The signal for the interaction is in the **second moment** (covariance structure) of the joint distribution.
- Carryover effects in the DGP inflate the BR means during off-drug periods, making them resemble on-drug means. This reduces the **differential** in the correlation structure between treatment states, weakening the signal that the analysis model detects.

This architecture is used in the Hendrickson et al.[4] publication code and the audited `R/` package in pmsimstats-ng.

A third possibility is that both channels operate at once: the biomarker could scale the mean drug effect (mean moderation) while simultaneously altering the biomarker-response correlation during active treatment (differential correlation). Such a dual-channel architecture, carrying independent weights on the two channels, recovers the two architectures above as its single-channel limits and is the natural framework when the operative mechanism is uncertain. Because it introduces a second free parameter and its power behavior warrants separate treatment, we develop it in a companion paper[6] and restrict the present paper to the contrast between mean and covariance moderation.

3.3 2.3 Mathematical Comparison

We begin the comparison by considering the expected outcome difference between two participants with biomarker values b_1 and b_2 at the same drug exposure level D under the mean-moderation architecture:

$$E[Y \mid b_1, D] - E[Y \mid b_2, D] = \beta_1 \cdot \beta_{bm} \cdot (b_1 - b_2) \cdot D$$

The absolute magnitude of this difference scales linearly with D and therefore shrinks during off-drug periods when D is replaced by an exposure-decayed value $D \cdot \text{carryover}$. We note, however, that the **ratio** of the drug effect between high-biomarker and low-biomarker participants is preserved at every level of exposure. The biomarker-treatment signal contracts in amplitude but does not lose its identifiability under carryover.

Under the covariance-moderation architecture, the analogous quantity is:

$$E[BR|b_1, D=1] - E[BR|b_2, D=1] = c_{bm} \cdot \frac{\sigma_{BR}}{\sigma_B} \cdot (b_1 - b_2)$$

During off-drug periods with carryover:

$$E[BR|b, D=0, t_{sd}] = \mu_{BR,\text{off}}(t_{sd}) + c_{bm} \cdot e^{-\lambda t_{sd}} \cdot \frac{\sigma_{BR}}{\sigma_B} \cdot (b - \mu_B)$$

Here the biomarker-dependent component decays exponentially with time off drug. As t_{sd} increases, the off-drug conditional expectation converges to $\mu_{BR,\text{off}}$ for all biomarker values, eliminating the differential signal entirely.

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4 3. Consequences for Statistical Power Under Carryover

4.1 3.1 Empirical demonstration

We ran identical simulation configurations under both architectures using the pmsimstats-ng framework, matching the DGP parameters as closely as possible between the two approaches. We note that the non-biologic response components (time-varying response, placebo-belief accumulation, and residual noise) are generated from the same multivariate normal draw with identical means, variances, within-factor AR(1) autocorrelation, and cross-factor correlations in both architectures. The two DGPs differ *only* in how the biomarker-to-biological-response (BM-BR) linkage is encoded: the covariance-moderation architecture embeds it in the off-diagonal c_{bm} entries of Σ, whereas the mean-moderation architecture omits those entries and adds a post-draw shift $c_{bm} \cdot B_i^* \cdot \sigma_{BR}$ to on-drug BR columns (see `19-mean-moderation-implementation-notes.tex` for the scaling derivation). The divergent carryover behavior reported below is therefore attributable to the BM-BR parameterization, not to any difference in how the placebo, time-varying, or residual-noise components are generated. The analysis model was the same in both cases: an `nlme::lme` fit with a `corCAR1` continuous-time autocorrelation structure and a single biomarker-by-drug interaction term, written `bm:Dbc` in R formula syntax.

The analysis-model drug exposure variable `Dbc` is the *continuous* analogue of the binary DGP indicator D_{it}. It equals 1 during on-drug periods and $\exp(-\lambda t_{sd})$ during off-drug periods, where t_{sd} is time since drug discontinuation and λ is the analysis-model carryover decay rate (typically derived from the drug's pharmacokinetic half-life as $\lambda = \ln 2/t_{1/2}$, in which case it matches the DGP decay rate of Section 2.2 by construction). Whereas D_{it} flips abruptly between 0 and 1, `Dbc` captures the smooth shrinkage of residual drug effect during the off-drug period. The interaction coefficient on `bm:Dbc` is therefore the test of whether the biomarker moderates the exposure-decayed drug effect.

We follow the ADEMP framework of Morris, White, and Crowther (2019): Aims, Data-generating mechanisms, Estimands, Methods, Performance measures.

Setup:

- Trial designs: CO (traditional crossover), Hybrid (N-of-1 Hybrid), OL+BDC (open-label with blinded discontinuation)
- Sample size: 35 participants allocated to each randomization path, so the total is $N = 70$ for the two-path CO and OL+BDC designs and $N = 140$ for the four-path Hybrid design. Throughout this compendium N denotes the total across paths; because this driver fixed the per-path count rather than the total, the three designs are not matched on N, and the cross-design comparison in Section 3.1 must be read with that in mind (see Notes and reproducibility)
- Biomarker correlation / moderation: $c_{bm} = 0.45$ / $\beta_{bm} = 0.45$
- DGP carryover half-life: $t_{1/2} \in \{0, 0.5, 1.0\}$ weeks
- Analysis: matched `Dbc` with same half-life
- Replicates: 1,000 per cell (the Monte Carlo standard error (MCSE) on a binomial power estimate near 0.5 is approximately $\sqrt{0.5 \cdot 0.5/1{,}000} \approx 0.016$)
- Implementation: 8-core `parallel::mclapply`; full factorial 2 architectures $\times$ 3 designs $\times$ 3 $t_{1/2}$ $\times$ 2 c_{bm} levels = 36 cells, 36,000 fits, 100% convergence

The 35-per-path choice is the smallest of the prazosin-PTSD- calibrated sample sizes used in the published methodological literature; it is a regime in which neither architecture saturates power at the no-carryover cells, so any carryover-induced erosion is mechanically visible. Type I error under the $c_{bm} = 0$ null cells sat in $[0.010, 0.074]$ across the 18 null cells. These rates below the nominal 0.05 are not architecture-specific; they reflect a known conservatism of the standard `corCAR1` Wald test, whose model-based standard error is over-stated by roughly 6 to 10 percent, characterized in a companion calibration study[8] together with a cluster-robust remedy. Being common to both architectures, the conservatism does not affect the architecture contrast.

Results under the mean-moderation architecture:

Design	$t_{1/2} = 0$	$t_{1/2} = 0.5$	$t_{1/2} = 1.0$
CO	0.739	0.739	0.725
Hybrid	0.992	0.987	0.978
OL+BDC	0.751	0.748	0.730

[9]

Under the mean-moderation architecture the carryover-induced power loss is modest in every design (1-3% relative).

- Relative loss to $t_{1/2} = 1$: 1.9% (CO), 1.4% (Hybrid), 2.8% (OL+BDC).
- Below the 8-13% range earlier exploratory runs suggested.
- Power is essentially intact under carryover.

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Power declines modestly under the mean-moderation architecture: relative loss from $t_{1/2} = 0$ to $t_{1/2} = 1$ is 1.9% (CO), 1.4% (Hybrid), and 2.8% (OL+BDC). We note that all three values fall well below the 8-13% relative-loss range suggested by earlier exploratory runs in this project for direct-mean-moderation DGPs, though they reproduce the qualitative finding of those runs, namely a modest erosion that leaves power essentially intact.

Results under the covariance-moderation architecture (differential correlation, MVN):

Design	$t_{1/2} = 0$	$t_{1/2} = 0.5$	$t_{1/2} = 1.0$
CO	0.745	0.754	0.723
Hybrid	0.995	0.980	0.944
OL+BDC	0.777	0.694	0.539

[10]

Under the covariance-moderation architecture the loss is much larger and strongly design-dependent, reaching 30.6% at OL+BDC but absent at CO.

- Relative loss: 3.0% (CO, not significant), 5.1% (Hybrid), 30.6% (OL+BDC) – about 11x the mean-moderation architecture loss.
- The CO cell fluctuates within MC error, consistent with no architecture-by-carryover interaction there.
- Below the 40-60% range earlier runs suggested, but confirming substantial, design-dependent erosion.

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Power declines more sharply under the covariance-moderation architecture, and most markedly so in the OL+BDC design: relative loss from $t_{1/2} = 0$ to $t_{1/2} = 1$ is 3.0% (CO; not significantly different from the mean-moderation architecture's CO loss), 5.1% (Hybrid), and **30.6% (OL+BDC)**. The CO cell shows a small, non-monotone fluctuation ($0.745 \to 0.754 \to 0.723$), which is within one MCSE and is consistent with the null finding of no significant architecture-by-carryover interaction in that design. The OL+BDC loss is approximately 11 times the matched mean-moderation OL+BDC loss (2.8%). It falls below the 40-60% relative-loss range suggested by earlier exploratory runs in this project for the MVN differential-correlation DGP, but it confirms the qualitative finding of those runs, namely a substantial and strongly design-dependent erosion.

We compare the carryover-induced power loss between architectures using a difference-of-differences z-statistic. Let $\Delta_X = \hat{\pi}_{X,0} - \hat{\pi}_{X,1}$ denote the within-architecture loss from $t_{1/2} = 0$ to $t_{1/2} = 1.0$ for architecture $X \in \{A, B\}$, estimated from $n = 1{,}000$ independent replicates at each cell. The B-versus-A gap is $\Delta_B - \Delta_A$, with standard error

$$\mathrm{SE} = \sqrt{\frac{\hat{\pi}_{B,0}(1-\hat{\pi}_{B,0})}{n} + \frac{\hat{\pi}_{B,1}(1-\hat{\pi}_{B,1})}{n} + \frac{\hat{\pi}_{A,0}(1-\hat{\pi}_{A,0})}{n} + \frac{\hat{\pi}_{A,1}(1-\hat{\pi}_{A,1})}{n}}$$

since the four proportions come from independent replicate sets. All tests are two-sided. At OL+BDC this gives $z = 7.64$ ($p < 10^{-13}$); at Hybrid, $z = 3.96$ ($p < 10^{-4}$); the CO contrast is not statistically distinguishable from zero ($z = 0.29$) because the CO design's within-subject AR(1) structure dominates the carryover-mediated correlation decay channel that the covariance-moderation architecture uses to encode the biomarker-treatment interaction.

Worked reading of the contrast. At OL+BDC, the within-architecture carryover loss is $\Delta_A = 0.751 - 0.730 = 0.021$ (2.8% relative) and $\Delta_B = 0.777 - 0.539 = 0.238$ (30.6%). The quantity of interest is the difference-of-differences $\Delta_B - \Delta_A = 0.217$ (about 22 percentage points), and the z-statistic above scales it by its standard error to give $z = 7.64$ ($p < 10^{-13}$): the architectures' carryover behavior differs well beyond Monte Carlo error. The same contrast is clearly positive for the Hybrid design and indistinguishable from zero for the crossover design.

4.2 3.2 Why the architectures diverge

We turn now to a mechanistic account of why the two architectures produce such different power losses under carryover.

[11]

Under the mean-moderation architecture, carryover only compresses the treatment contrast; the biomarker signal-to-noise ratio is nearly preserved.

- Carryover shrinks `Dbc` and inflates off-drug BR means (shared with the covariance-moderation architecture).
- But β_{bm} is a fixed mean parameter with drug-state- independent noise.
- So the interaction variance falls but the signal-to-noise ratio degrades only modestly.

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Under the mean-moderation architecture, carryover reduces the magnitude of the drug exposure variable seen by the analysis model (`Dbc` drops below 1 during the off-drug period), and the mean BR at off-drug timepoints is inflated by the residual drug effect exactly as in the covariance-moderation architecture. Both mechanisms compress the between-state treatment contrast and are shared by the two architectures. What the mean-moderation architecture lacks is the second channel of signal loss: the biomarker moderation coefficient β_{bm} is a fixed population parameter, the biomarker enters the mean structure directly rather than through a correlation, and the noise structure is independent of drug state. The biomarker-by-drug interaction term therefore has reduced variance (because `Dbc` has less contrast between drug states), but the signal-to-noise ratio degrades only modestly.

Under the covariance-moderation architecture, carryover operates on two levels simultaneously:

1. **Mean blurring**: The BR means during off-drug periods are inflated by residual carryover, making them resemble on-drug BR means. This reduces the treatment contrast that the analysis model uses to distinguish drug effect from no-drug. This channel is shared with the mean-moderation architecture.
2. **Correlation erosion**: The BM-BR correlation during off-drug periods decays exponentially with t_{sd}. This is not just a statistical modeling choice; it is a structural consequence of the DGP. The correlation between biomarker and BR reflects the degree to which the drug effect is active; as the drug washes out, the correlation weakens because the drug-mediated component

of BR variance shrinks relative to the drug-independent component. This channel is *unique* to the covariance-moderation architecture, because only here does the biomarker signal live in the second moment of the joint distribution. Under the mean-moderation architecture the drug-mediated and drug-independent components of BR variance move in tandem (both are scaled by the same `Dbc`-weighted drug effect), so their ratio, and hence the identifiability of the interaction, is preserved even as amplitudes shrink.

[12]

The covariance-moderation architecture's interaction signal is attacked from both sides, which the mean-moderation architecture escapes – hence the divergence.

- The `bm:Dbc` coefficient depends on the covariance of B_i with `Dbc`-weighted outcomes.
- Under B both the `Dbc` contrast and the BM-BR correlation decay together.
- Under A the second attack is absent, so power is largely preserved.

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The analysis model's interaction coefficient (the coefficient on `bm:Dbc`) depends on the covariance between B_i and `Dbc`-weighted outcomes. Under the covariance-moderation architecture, this covariance is being attacked from both sides: the treatment contrast in `Dbc` is compressed, and the BM-BR correlation that generates the covariance signal is simultaneously decaying. That the mean-moderation architecture escapes the second attack entirely accounts for the divergence in power loss documented above.

4.3 3.3 Robustness check at $N = 140$

[13]

An $N = 140$ confirmation run checks that the architecture ordering survives larger samples.

- Section 3.1 puts $N = 70$ on this design, where the loss is mechanically visible (no ceiling saturation).
- The check runs 12 cells at $N = 140$ (70 per path), 800 replicates each, on the two-path OL+BDC design.
- 100% convergence across 9,600 fits.

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The Section~3.1 production results put $N = 70$ on this design, where neither architecture saturates power and the carryover-induced loss is mechanically visible. We conducted a confirmation run at $N = 140$ to check whether the qualitative ordering survives the saturation that comes with the larger sample size. An 800-replicate-per-cell Monte Carlo run was executed at $N = 140$ across 12 cells (architecture $\in \{\text{MVN}, \text{mean moderation}\} \times c_{bm} \in \{0, 0.45\} \times t_{1/2} \in \{0, 0.5, 1.0\}$ weeks), all on the OL+BDC design, using 8-core `parallel::mclapply`. Wall-clock 483~s; convergence 100% across 9{,}600 fits.

At $c_{bm} = 0.45$:

[14]

At $N = 140$ the B-versus-A ordering persists, though absolute magnitudes are compressed by ceiling saturation.

Architecture	$t_{1/2} = 0$	$t_{1/2} = 0.5$	$t_{1/2} = 1.0$
MVN (B)	0.978	0.945	0.885
Mean moderation (A)	0.976	0.973	0.959

- The covariance-moderation architecture loses 9.3 points ($z = 7.4$); the mean-moderation architecture 1.7 points ($z = 2.0$).
- Both sit within 2-5 points of the ceiling at $t_{1/2} = 0$, truncating the absorbable loss.
- The Section 3.1 results remain the canonical reference for the narrative magnitude.

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Within-architecture losses from $t_{1/2} = 0$ to $t_{1/2} = 1$ are 9.3 percentage points (the covariance-moderation architecture; two-proportion z-test on the paired cells, $z = 7.4$, $p < 10^{-12}$) and 1.7 percentage points (the mean-moderation architecture; $z = 2.0$, $p \approx 0.05$), where each z-statistic uses the formula $(\hat{\pi}_0 - \hat{\pi}_1)/\sqrt{\hat{\pi}_0(1-\hat{\pi}_0)/n + \hat{\pi}_1(1-\hat{\pi}_1)/n}$ with $n = 800$ replicates per cell. The qualitative ordering matches Section~3.1's result. The absolute magnitudes are compressed because $N = 140$ at the prazosin-PTSD calibration places both architectures within 2-5 percentage points of the unit ceiling at $t_{1/2} = 0$, mechanically truncating the loss either architecture's curve can absorb. We note that reading the narrative loss-magnitude claim against the $N = 140$ robustness numbers (3% and 9% relative loss for the covariance-moderation architecture versus 0.4% and 1.8% for the mean-moderation architecture) requires acknowledging this saturation effect; the production results in Section~3.1 are the canonical reference for the narrative magnitude.

Type I error. Across all 18 null cells of the Section~3.1 production run, Type~I error sat in $[0.010, 0.074]$, with mean 0.043 and median 0.042. We observe no architecture-specific inflation. The uniformly sub-nominal rates are consistent with the `corCAR1` working-covariance miscalibration characterized in the companion calibration study[8]; because it affects both architectures alike, it is common-mode with respect to the architecture contrast. The covariance-moderation architecture's $t_{1/2} = 1.0$ OL+BDC null ($p = 0.010$, slightly conservative) is the smallest cell; the corresponding alternative ($p = 0.539$) is the most informative cell of the Section~3.1 narrative.

Estimator bias. Mean estimated $\hat{\beta}_{bm:D_{bc}}$ at $c_{bm} = 0.45$ ranged from -0.231 to -0.281 across the 18 alternative cells, with the $N = 70$ mean-moderation OL+BDC cell at $t_{1/2} = 1.0$ producing the largest absolute mean (-0.281); the covariance-moderation architecture cells maintained mean near -0.234 across all $t_{1/2}$ values, consistent with that architecture's information channel being eroded through correlation decay rather than through coefficient drift. Empirical SE of $\hat{\beta}_{bm:D_{bc}}$ runs 0.05 to 0.10 across cells; the within-replicate model SE matches at scale, indicating nominal-95% Wald coverage holds across the grid.

---

5 4. Which recipe matches which biology?

The choice of architecture is really a choice about *why* the biomarker predicts response, so picking one is a biological statement, not a coding preference.

[15]

Use the mean-moderation architecture when the biomarker mechanically sets the size of the drug effect.

- The biomarker fixes effective dose or receptor occupancy, so it scales the effect proportionally.
- Doubling the biomarker roughly doubles the effect at every timepoint.
- Carryover keeps the same proportion in the off-drug period, so power is barely touched.

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Use the mean-moderation architecture when the biomarker sets the size of the drug effect. If a patient's biomarker mechanically determines how much drug effect they get – for example, kidney function (eGFR) controlling how fast a drug is cleared, or receptor density controlling how strongly a drug acts – then doubling the biomarker roughly doubles the effect at every timepoint. Carryover does not break this: the leftover effect during the off-drug period keeps the same proportion, so power is barely touched.

Typical examples:

- **Drug clearance.** Kidney function (eGFR) sets how fast a renally-cleared drug leaves the body, so it directly scales the steady-state drug level and hence the effect.
- **Receptor density.** If a biomarker measures how many target receptors a patient has, more receptors means proportionally more drug effect.
- **Metabolizer genotype.** A "poor metabolizer" of a prodrug activates less of it, getting a proportionally smaller effect.

  [16]

Use the covariance-moderation architecture when the biomarker is only a proxy for a hidden responder state.

- The biomarker imperfectly indexes the real cause, so identical biomarkers can respond differently.
- It predicts response only while the drug is acting.
- In the off-drug period the correlation fades and power drops sharply, as in Section 3.

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Use the covariance-moderation architecture when the biomarker only flags who is likely to respond. Sometimes the biomarker is just a *proxy* for some hidden "responder" state. Two patients with the same biomarker can respond very differently, because the biomarker imperfectly indexes the real cause (a disease subtype, an inflammatory state, a genetic background). Here the biomarker predicts response *only while the drug is acting*; once the drug washes out, the correlation that carried the signal fades, and power drops sharply – exactly the covariance-moderation architecture behavior we saw in Section 3.

Typical examples:

- **Blood pressure in PTSD.** High resting blood pressure may flag a noradrenergic PTSD subtype likely to respond to prazosin, but it does not *cause* the response; it is a noisy marker of the real underlying state.

- **Inflammatory markers.** High C-reactive protein (CRP) or interleukin-6 (IL-6) may predict response to anti-inflammatory augmentation of an antidepressant, but only on average – many high-CRP patients still do not respond.

- **Polygenic risk scores.** These predict treatment response while explaining only a fraction of the variance; the rest is unmeasured biology.

(A fully faithful model of this "hidden responder class" idea would be a finite mixture; the covariance-moderation architecture is a convenient correlation-level approximation, adequate for power calculations but not a claim about the true mechanism. A companion paper[5] develops the mixture and latent-class formulations, deriving when a single multivariate normal reproduces a two-component mixture's first two moments and cataloguing where it fails. Senn[9] also cautions that apparent biomarker-treatment effects are sometimes just variance artifacts, so the covariance-moderation architecture's signal may be smaller than assumed.)

[17]

The motivating prazosin/PTSD case is a covariance-moderation story, so its carryover power loss is a clinically relevant warning.

- Biomarker is resting blood pressure, a proxy for noradrenergic tone, not a driver of drug levels.
- Designs with long off-drug periods may lose more power than a mean-moderation simulation predicts.
- If blood pressure also acts through the mean channel, the dual-channel architecture of the companion paper applies.

rgt to complete.

The motivating example. The pmsimstats software was built for a prazosin/PTSD trial[4,10] where the biomarker is resting blood pressure and the outcome is a PTSD symptom score. Blood pressure is thought to be a *proxy* for noradrenergic tone, not a direct driver of drug levels – a covariance-moderation story. So the large carryover-driven power loss we find under the covariance-moderation architecture is the clinically relevant warning: designs with long off-drug periods may lose more power to detect this biomarker than a simpler mean-moderation simulation would suggest. Should blood pressure turn out to act through the mean channel as well as the correlation channel, the dual-channel architecture of the companion paper[6] is the appropriate model; because it estimates two independent parameters, it calls for independent evidence on each.

---

6 5. Implications for Simulation Study Design

6.1 5.1 Choosing the appropriate architecture

The choice of DGP architecture should be guided by the biological mechanism hypothesized for the biomarker-treatment interaction:

[18]

Criterion	Mean moderation	Covariance moderation
Biomarker role	Causal mediator	Statistical predictor
Mechanism	Deterministic scaling	Probabilistic association
Signal location	Mean structure	Covariance structure
Free parameters	1 (c_{bm})	1 (c_{bm})
Carryover impact	Modest (1–3%)	Design-dependent, up to 31%
Appropriate when	Biomarker determines dose or PK	Biomarker indexes a latent subtype

When the mechanism is ambiguous, use the covariance-moderation architecture as the conservative default.

- The covariance-moderation architecture is the safer default for design decisions.
- Its larger carryover sensitivity keeps power estimates from being optimistic.
- A dual-channel sweep across mixing weights is available in the companion paper.

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When the operative mechanism is unknown, the covariance-moderation architecture is the conservative default for trial-design decisions: its larger carryover sensitivity keeps power estimates from being optimistic relative to the mean-moderation architecture. A dual-channel architecture that mixes the two channels, together with a corresponding sensitivity sweep over mixing weights, is developed in the companion paper[6].

6.2 5.2 Reporting requirements

Simulation studies evaluating biomarker-moderated treatment effects should explicitly state:

1. Whether the biomarker-treatment interaction is generated through mean moderation, differential correlation, or both channels simultaneously.
2. The biological rationale for the chosen mechanism, with independent evidence supporting each channel that is activated.
3. Whether the carryover model (if present) operates on the interaction signal consistently with the chosen architecture.
4. Sensitivity analyses under alternative architectures, if the biological mechanism is uncertain.

[19]

These requirements specialize the ADEMP template; the architecture choice is its data-generating (D) element.

- ADEMP fixes Aims, Data-generating mechanisms, Estimands, Methods, and Performance measures.
- The DGP-architecture choice is the D element.
- Our reporting rules are ADEMP specialized to architecture-sensitive biomarker simulations.

rgt to complete.

> The reporting framework above is consistent with the broader ADEMP template of Morris, White, and Crowther[11], which prescribes that simulation studies fix five elements before production runs: the **Aims** of the study, the **Data-generating mechanisms**, the **Estimands** (the inferential targets), the **Methods** (analysis specifications compared), and the **Performance measures** (power, bias, coverage, Type I error, and MCSEs of each). The DGP-architecture choice this paper concerns is the **D** element of an ADEMP specification; the reporting requirements above are the specialization of the broader ADEMP framework to the architecture-sensitive class of biomarker-validation simulations.

The **primary estimand** for this paper's simulations is the biomarker-treatment interaction coefficient $\beta_{bm:D}$ in the linear-mixed analysis model `Sx ~ bm + t + Dbc + bm:Dbc` with random intercept and AR(1) residual correlation. The reported **performance measures** are: (i) power for $\beta_{bm:D}$ at $\alpha = 0.05$, with MCSE $\sqrt{\hat{\pi}(1-\hat{\pi})/n_{\text{reps}}}$; (ii) Type I error under $c_{bm} = 0$, with the same MCSE; (iii) bias of $\hat{\beta}_{bm:D}$ relative to the calibrated true value at each parameter cell; (iv) the empirical MCSE of $\hat{\beta}_{bm:D}$ across replicates; and (v) the convergence rate of the linear-mixed fit. Sample size, biomarker effect size, and carryover half-life vary across the parameter grid; the number of replicates per cell is calibrated to achieve MCSE below 0.02 for power at $\hat{\pi} = 0.5$, which corresponds to $n_{\text{reps}} \geq 625$. The simulations reported here use $n_{\text{reps}} = 1000$ throughout.

6.3 5.3 When the choice is uncertain

If the biological mechanism is ambiguous, as it often is in early biomarker development, we suggest investigators take the following steps:

1. Run the simulation under both architectures and report the range of power estimates.
2. The covariance-moderation architecture produces more conservative (lower) power estimates under carryover, making it the safer default for trial design decisions.
3. Design the trial to minimize carryover (adequate washout periods) to reduce sensitivity to the DGP architecture choice.

---

7 6. Can a smarter analysis recover the lost power?

None of the strategies below was tested by simulation here; the companion paper on carryover-mitigation strategies does that. This section just lists the options and their intuition.

> [20]
>
> **The covariance-moderation architecture's loss has two sources; only the variance-compression part is recoverable, by down-weighting low-signal off-drug points.**
>
> - Source 1: the drug exposure variable's variance shrinks (recoverable).
> - Source 2: the biomarker-response correlation genuinely fades off-drug (lost information).
> - Methods that down-weight or drop off-drug points can help with the first.
>
> *rgt to complete.*

Under the covariance-moderation architecture the power loss has two sources: the drug exposure variable's variance shrinks, and the biomarker-response correlation genuinely fades off-drug. The second is lost information that no model can recover. But because off-drug points carry less signal, methods that *down-weight or drop* them can help.

The natural candidates are:

- **Use on-drug points only.** Every kept point has the full correlation, but you lose the off-drug reference and sample size; best for designs with a long on-drug phase, worst for crossover.
- **Weight by drug exposure.** Keep all points but weight each by its expected residual drug effect, $w_t = e^{-\lambda t_{sd}}$, so high-signal points count more. Easy to do with the `weights` argument in `nlme::lme`.
- **Within-subject contrast.** Reduce each patient to one number, (mean on-drug) minus (mean off-drug), then regress that on the biomarker: $\bar{Y}_{i,\text{on}} - \bar{Y}_{i,\text{off}} = \alpha + \gamma B_i + \epsilon_i$. Simple and robust, at some cost in efficiency.
- **Two-stage random slopes.** First estimate each patient's drug effect u_{1i} from a random-slopes model, then regress those on the biomarker, $\hat{u}_{1i} = \delta_0 + \delta_1 B_i + \eta_i$.
- **Drop the most contaminated early off-drug points**, or **fix it in the design** with longer washouts or more on-drug time.

7.1 6.1 Qualitative comparison

Table~1 summarizes the strategies along three dimensions: whether they discard observations, implementation complexity, and the expected direction of benefit based on the mechanistic account in Section 3.2. All entries in the 'Hypothesized benefit' column are qualitative; no simulation evidence is presented here, and each strategy should be evaluated empirically before use in trial planning.

A systematic simulation comparison of these strategies under the covariance-moderation architecture, analogous to the factorial comparison of analysis models in Section 3, would identify which approach best recovers the power lost to carryover for each trial design. We regard this comparison as important future work and note that the on-drug-only restriction and the weighted analysis are the most tractable candidates for a first empirical evaluation.

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8 7. How this fits the wider literature

[21]

The mean-moderation architecture is near-universal in the literature, so published power estimates may be optimistic for the covariance-moderation architecture biomarkers.

- Simon, Freidlin-Korn, Renfro-Sargent, and standard N-of-1 studies all put the effect in the mean.
- Even N-of-1 carryover work sharing Hendrickson's authors uses the mean-moderation architecture, so B looks like a one-off.

Table 1: Qualitative summary of analysis strategies for recovering power under the covariance-moderation architecture carryover. All benefit assessments are mechanistic hypotheses, not simulation-supported estimates.

Strategy	Discards data	Complexity	Hypothesized benefit
On-drug only	Yes	Low	Eliminates correlation erosion; forfeits off-drug reference (best for OL/Hybrid)
Weighted	No	Low	Reduces erosion damage proportionally; depends on weight calibration
Within-subject contrast	Aggregates	Low	Robust to carryover misspecification; less efficient than the linear mixed-effects (LME) model
Two-stage random slopes	No	Moderate	Unknown; separates estimation stages; robustness unverified
Exclude early off-drug	Some	Low	Removes most contaminated observations; improves contrast clarity
Design modification	N/A	N/A	Addresses root cause; cost is longer trial or reduced carryover estimation

- Most published power figures may be optimistic for genuinely the covariance-moderation architecture biomarkers.

rgt to complete.

Almost every trial-simulation framework uses the mean-moderation architecture: Simon[2], Freidlin and Korn[3], Renfro and Sargent[12], and the standard N-of-1 simulation studies all put the biomarker effect in the mean. Even methodological work on N-of-1 carryover that shares authors with Hendrickson et al.[4] uses the mean-moderation architecture, which suggests Hendrickson's differential-correlation choice was a one-off rather than a convention. The practical worry is that most published power estimates may be *optimistic* for biomarkers that really behave like the covariance-moderation architecture, because they ignore the extra carryover-driven loss we have demonstrated.

[22]

The distinction matters most in treatment-switching designs; when the mechanism is unknown, compute power both ways and use the lower number.

- Treatment-switching designs (crossover, N-of-1, basket) put patients through off-drug phases where B's correlation decays.
- Parallel-group designs have no off-drug phase, so the architecture choice does not matter.
- This is a DGP question, separate from the analysis-model guidance of the PATH statement.

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The distinction matters most exactly in the designs precision medicine cares about: any *treatment-switching* design (crossover, N-of-1 hybrid, basket/platform) puts patients through off-drug phases where the covariance-moderation architecture correlation decays. Parallel-group designs, with no off-drug phase, are immune and give similar numbers under either architecture. The simple advice when the mechanism is unknown: run the power calculation *both ways* and use the lower (the covariance-moderation architecture) number for planning. This is a data-generating-process question, separate from – and not answered by – the analysis-model guidance in the PATH statement[7].

To see where this bites in practice, it helps to sort designs by how much off-drug time each patient experiences:

- **Treatment-switching designs** (crossover, N-of-1 hybrid, basket and platform trials) deliberately cycle each patient on and off drug. Under the covariance-moderation architecture every off-drug period erodes the correlation, so the interaction signal in later periods is weaker than in the first – the power loss of Section 3 applies directly.
- **Parallel-group enrichment designs** keep each patient on one treatment throughout. There is no off-drug phase for the correlation to decay in, so the mean-moderation and covariance-moderation architectures give similar power. The architecture choice simply does not matter here.
- **Adaptive enrichment trials** decide mid-trial whom to keep enrolling, using interim power calculations written in the the mean-moderation architecture idiom. If the truth is closer to the covariance-moderation architecture and patients pass through washouts, the real power is lower than the rule assumes, which can trigger premature stopping or an over-narrow eligible population.

The pattern is worth stating plainly: the architecture choice matters *most* in exactly the within-subject designs that precision medicine finds most attractive, and *least* in the parallel-group designs where it is often (wrongly) assumed to be harmless.

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9 8. Conclusions

In conclusion, six points are to be emphasized.

1. The choice between direct mean moderation (the mean-moderation architecture) and differential correlation (the covariance-moderation architecture) for generating biomarker-treatment interactions in clinical trial simulations has substantive consequences for power estimates under carryover.
2. Under the mean-moderation architecture, carryover reduces power only modestly, by 1 to 3% in relative terms across the three designs examined, because the proportional biomarker-drug relationship is preserved during the off-drug period. Under the covariance-moderation architecture the loss is strongly design-dependent: it remains modest in the crossover and hybrid designs (3 to 5%) but reaches roughly 31% in the open-label OL+BDC design, because the differential-correlation signal erodes as the drug effect wanes.

3. The choice should be guided by the hypothesized biological mechanism: the mean-moderation architecture for biomarkers that directly determine drug pharmacokinetics or pharmacodynamics; the covariance-moderation architecture for biomarkers that statistically predict treatment response through latent subtype membership.
4. For the prazosin-PTSD application with blood pressure as the predictive biomarker, the covariance-moderation architecture (MVN differential correlation) is the more appropriate model, and the resulting power sensitivity to carryover should inform trial design decisions regarding off-drug period duration.
5. The existing clinical trial simulation literature uses the mean-moderation architecture almost exclusively. The Hendrickson et al. (2020) MVN approach is, to the best of our knowledge, unique in the N-of-1 trial context. This means that published power estimates for biomarker-moderated crossover and N-of-1 designs may be systematically optimistic for biomarkers that operate through a correlation-based (the covariance-moderation architecture) mechanism, because they do not account for the carryover sensitivity documented here.
6. Simulation studies for biomarker-moderated treatment effects should explicitly report their DGP architecture and consider sensitivity analyses under the alternative when the biological mechanism is uncertain.

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10 Notes and reproducibility

The authors declare no conflicts of interest and received no specific funding. Code and data are in the pmsimstats-ng repository (drivers under `analysis/scripts/`, outputs under `analysis/data/`), with package versions pinned in `renv.lock`; the document renders under R~4.6.0.

Sample size, and one limitation it carries. N denotes the total number of participants across all randomization paths. The driver behind Sections 3.1 and 3.3 fixed the per-path count instead: 35 per path in Section~3.1 and 70 per path in Section~3.3. Because paths are concatenated before analysis, the totals are $N = 70$ (two-path CO and OL+BDC) and $N = 140$ (four-path Hybrid) in Section~3.1, and $N = 140$ and $N = 280$ respectively in Section~3.3.

The designs are therefore not matched on N, and the Hybrid row of each Section~3.1 results table rests on twice the sample of the other two. Cross-design differences in absolute power, such as the Hybrid advantage in those tables, are consequently confounded with sample size and should not be read as design effects. The claims this paper actually makes are within-design contrasts between the two architectures, and those hold N fixed within each design, so they are unaffected. A matched re-run at equal totals would be needed before the designs themselves could be ranked.

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